Introductory activity

Activity 1 · Fine weather, rain 40 min . In a simplified model (fictitious data), the weather on a given day is either fine (S) or rainy (P), and depends only on the weather the day before: after a fine day, it is fine with probability 0.8; after a rainy day, it is fine with probability 0.4. Day 0 is fine. Let s_n and p_n be the probabilities that day n is fine or rainy.

1. Represent the situation by a diagram: two vertices S and P, an arrow for each possible change, carrying its probability.
2. Using a tree, compute s_1, p_1, then s_2, p_2.
3. Justify that $s_{n+1} = 0.8s_n + 0.4p_n$ and $p_{n+1} = 0.2s_n + 0.6p_n$.
4. Let $\pi_n = \begin{pmatrix} s_n & p_n \end{pmatrix}$ and $T = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$. Check that $\pi_{n+1} = \pi_n T$. Compute π_{10} with a calculator. Start again with a rainy day 0. What do you notice?
5. Find the distribution $\pi = \begin{pmatrix} a & b \end{pmatrix}$, with $a + b = 1$, that satisfies $\pi T = \pi$. Compare it with the observations of question 4.

Observing is not proving: the properties conjectured here are established in the course.